

MODDABLE CHESS · VARIANT 83 OF 86

Toroidal Chess

8×8

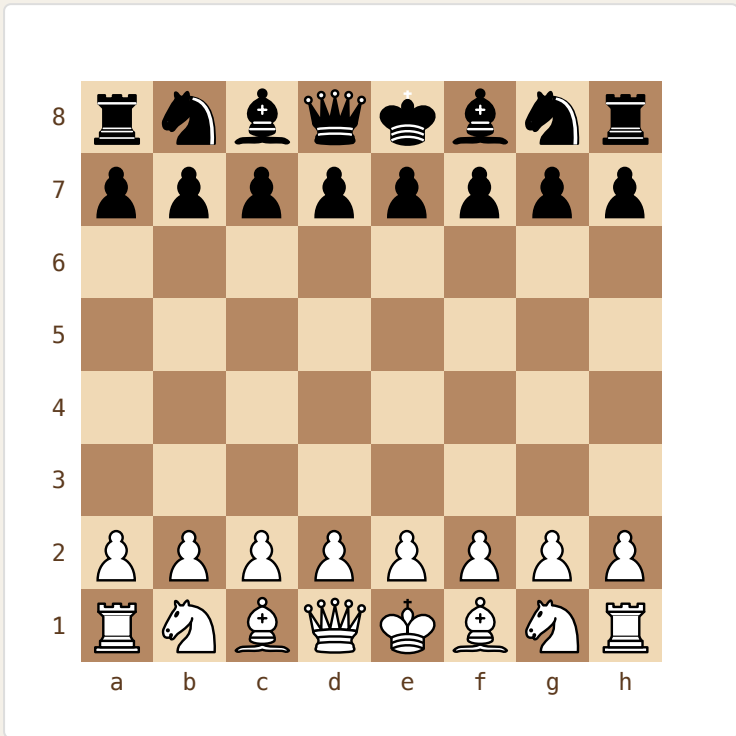
BOARD

2

PLAYERS

TOROIDAL CHESS

The board wraps in both directions: files connect left-to-right and ranks connect top-to-bottom, forming a torus. There are no edges.



Toroidal Chess – starting position

SETUP

Board: Standard 8×8 (wraps both axes).

Setup: Standard chess setup.

FEN: rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/RNBQKBNR w - - 0 1

RULES

- ♦ The a-file connects to the h-file (horizontal wrap).
- ♦ Rank 1 connects to rank 8 (vertical wrap).
- ♦ Rooks can slide in any direction and wrap around indefinitely until blocked.
- ♦ Bishops slide diagonally and wrap; a bishop can return to its starting square by sliding far enough.
- ♦ Queens combine both wrapping movements.
- ♦ Knights jump normally but the connectivity means all squares are reachable.
- ♦ No castling (no meaningful concept of king safety with wrapping ranks).
- ♦ No en passant.
- ♦ Pawns move “forward” relative to their colour (White toward rank 8, wrapping to rank 1; Black toward rank 1, wrapping to rank 8).

WIN CONDITION

Checkmate the opponent's King.

STRATEGY

On a torus, there are no safe corners or edges; every square is equally central. Bishops can reach every square of their colour (not just 7 diagonals but infinitely wrapping ones). Rooks threaten ranks and files that circle back, making them nearly impossible to evade. The opening concept of “King safety” is meaningless; positional play dominates.

ATTRIBUTION

Mathematical chess variant based on toroidal geometry. Public domain.